

SOUMEN MUKHERJEE

Lead Software Engineer | Java Backend | Distributed Systems | AWS & Cloud-Native
Bengaluru, India | +91-9564072741 | me.soumen02@gmail.com | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Lead Software Engineer with 8+ years of experience designing and delivering high-availability, fault-tolerant backend systems using Java, Spring Boot, and AWS. Deep expertise in microservices architecture, event-driven systems, REST API design, and distributed workflows at scale. Proven track record of leading cross-functional engineering initiatives, driving performance optimisations, and mentoring teams in agile, cloud-native environments. Strong command of SOLID principles, design patterns, multithreading, and production observability. Experienced in platform engineering at scale — across notification, healthcare, and government-grade systems.

TECHNICAL SKILLS

Languages	Java (8, 21), JavaScript
Backend	Spring Boot, Microservices, REST API Design, System Design, Distributed Systems, Event-Driven Architecture, SOLID Principles, Design Patterns, Multithreading & Concurrency
Cloud & DevOps	AWS (Lambda, EC2, S3, DynamoDB, SQS, SNS, CloudWatch), Docker, Kubernetes, CI/CD, Jenkins, Argo CD, Git, GitHub Actions
Messaging	Apache Kafka (KRaft), AWS SQS & SNS, Pub-Sub, Retry & DLQ patterns, Idempotency
Databases	PostgreSQL, DynamoDB, Oracle SQL, MySQL, Redis, JPA / Hibernate
Observability	Splunk, Grafana, Prometheus, AWS CloudWatch
Testing	JUnit, Mockito, Karate (API testing)
Frontend (Basic)	HTML, JavaScript, CSS, React.js

PROFESSIONAL EXPERIENCE

Associate Technical Architect | [Tavant Technologies \(Client: Booking.com\)](#) May 2026 – Present
Payout Config Change Coordinator | Java 21, Spring Boot, Kafka, AWS, MySQL, Prometheus (Mimir), OpenTelemetry, Grafana, Docker, Kubernetes, Harness

Graphite → Mimir

Metrics platform migrated

2

Grafana dashboards migrated

37

Alerts migrated

- Migrated production metrics for the payment-config-service (Spring Boot) from Graphite to Prometheus (Mimir), coordinating with the Prometheus platform team, and migrated all associated Grafana dashboards and alerts.
- Instrumented the service with OpenTelemetry and added the corresponding Harness pipeline config, introducing calculated SLO, min, max, and average latency metrics for every API call; deployed and verified in production.
- Added observability for the partner payout-method switch flow (VCC ⇌ Bank Transfer – NET/GROSS), introducing new metrics to capture downstream Bizent service call failures and “no asset payout relation found” (404) cases.
- Building an integration test suite for the asset payout-method switch flow for execution in the CI pipeline (in progress).
- Contributing to System Integration Testing (SIT) for BHFS Release 5.3 alongside ongoing feature work.

Lead Software Engineer | [Dexian India Technologies Pvt. Ltd. \(Client: Intuit\)](#) Apr 2024 – Apr 2026
One Intuit Notification Platform (OINP) | Java 21, Spring Boot, Kafka, AWS SQS/SNS, DynamoDB, PostgreSQL, Lambda, S3, Docker, Kubernetes, Argo CD, Splunk

2B+

Notifications / year

7d → 30m

Localisation SLA

400 TPS

Peak load validated

30+

Security fixes shipped

- Contributed to backend development of the One Intuit Notification Platform — a centralised, multi-channel notification system (Email, SMS, Push, Voice, VOIP, Tray) processing 2B+ notifications annually across Intuit's suite of products.
- Led end-to-end delivery of the Localisation framework of notification templates supporting 30+ locales; redesigned the pipeline using async AWS SNS/SQS, DynamoDB, and S3 with retry, DLQ, and CloudWatch alerting — cutting SLA from 7 days (manual) to 30 minutes.
- Validated new async architecture under a 400 TPS load test (TP99 ~3 s), leading the DC Pipeline deprecation and migration to the new event-driven design using Kafka.
- Engineered scalable microservices following domain-driven design and cloud-native principles; designed REST APIs and data models for high-availability, low-latency notification delivery.
- Led JDK upgrade of 5 production services from Java 8 to Java 21; identified and resolved 30+ security vulnerabilities across the platform.
- Built a reusable Claude Code skill enabling delivery of 4 new notification channels (Push, ICE, Braze, VOIP) in 10 days against a prior estimate of 1 week per channel — recognised for excellence by a Senior Staff Software Engineer and Senior Manager.
- Implemented production observability — structured logging, distributed tracing, and alerting — using Splunk and CloudWatch across all owned services.

Software Engineer II | [Oracle Cerner Healthcare Solutions India Pvt. Ltd.](#) Jun 2020 – Apr 2024
Bedrock — Healthcare Reference Data Platform | Java 8, Spring Boot, Oracle SQL, AWS (EC2, Lambda, DynamoDB), Jenkins, CloudWatch, Splunk

~4s TP99	2s → ms	5+	~4 yrs
Virtual View latency	App launch time	Microservices owned	Tenure at Oracle

- Senior backend engineer on Bedrock, Oracle Cerner's enterprise Reference Data Management platform enabling healthcare professionals to configure and access clinical reference data at scale.
- Designed and delivered Virtual View end-to-end — a new microservice for report reference data configuration — achieving a TP99 latency of ~4 seconds under production load.
- Optimised application cold-start performance via lazy loading and service refactoring, reducing launch time from ~2 seconds to milliseconds.
- Owned design, development, and maintenance of 5+ microservices on AWS (EC2, Lambda, DynamoDB) with Jenkins CI/CD and Splunk/CloudWatch observability.
- Led production incident response, maintaining stakeholder alignment on impact and resolution; recognised with 'Coach of the Quarter' award for technical ownership and knowledge transfer to new joiners.
- Participated in architectural discussions and code reviews focused on scalability, reliability, and performance in a regulated enterprise environment.

Software Engineer | [Allegis Services Pvt. Ltd. \(TEKsystems\) \(Client: Oracle Cerner\)](#) Jan 2020 – Jun 2020
Java 8, Spring Boot, Oracle SQL, AWS, Jenkins, Splunk

- Contributed to backend development of the Bedrock platform; delivered Spring Boot microservices and core Java APIs in a contract capacity at Oracle Cerner.
- Converted to a full-time Software Engineer II role at Oracle Cerner within 5 months due to strong execution and project impact.

Software Engineer | [Southern Phone Company Limited](#) Dec 2019 – Jan 2020
Java 8, Spring Boot, REST APIs, MySQL

- Built REST APIs for the customer inquiry module of the company's internal CRM platform — handling inquiry type and customer type metadata used across the inquiry management workflow.

Software Engineer | [Velocis Systems Pvt. Ltd. \(Client: National Informatics Centre\)](#) Feb 2019 – Nov 2019
TrackChild 3.0 — National Child Tracking Platform | Java 8, Spring Boot, PostgreSQL, Docker, Jenkins, REST APIs

Govt. (MeitY)	2.0 → 3.0	36	3
Platform	Migration led	States/UTs mapped	Modules built

- Led backend development and migration from TrackChild 2.0 to 3.0 — a national-scale government platform for locating missing children and matching them with orphaned children using face recognition.
- Designed and implemented the Feedback Module and CWC (Child Welfare Committee) stakeholder login, mapping all CWC entities across India's 36 states/UTs with dashboard and authentication flows.
- Optimised PostgreSQL queries, improved API response times, and containerised services using Docker with Jenkins CI/CD.

Junior Software Developer (Freelance) | Icode Solutions (Client: KRE Properties) Jan 2018 – Jan 2019

Java, Spring Boot, JavaScript, jQuery, CSS, MySQL

- Built a CRM platform for KRE Properties as a freelance developer — backend services in Java and Spring Boot, with the frontend built using JavaScript, jQuery, and CSS.

RECOGNITION & AWARDS

Excellence Award — Claude Code & Faster Delivery | Intuit, 2025

Recognised by a Senior Staff Software Engineer and Senior Manager for championing Claude Code adoption and building reusable engineering skills that delivered 4 new notification channels in 10 days against a 4-week estimate.

Coach of the Quarter | Oracle Cerner

Business-unit award for strong technical ownership, production incident leadership, and knowledge transfer to new joiners.

Contract to Full-Time in 5 Months | Oracle Cerner, 2020

Converted from a contract engagement to a full-time Software Engineer II role at Oracle Cerner within 5 months, based on performance and project impact.

WEEKEND PROJECTS — TRAILS LABS

Money Trails — Personal finance tracker — <https://money.trails.click>

- End-to-end microservices platform (9 services) for Indian investors: parses CAMS/KFintech/NSDL CAS PDFs, tracks mutual fund holdings, computes XIRR returns, and sends maturity and goal-based alerts.
- Event-driven architecture — all cross-service communication via Apache Kafka (3-broker KRaft cluster, self-hosted on TrueNAS); no direct service-to-service calls.
- Persistence: PostgreSQL (5 instances, one per service), MongoDB (CAS raw documents + notification history), Redis (JWT blacklist, rate limiting, NAV cache).
- Nightly batch pipeline: Scheduler + Spring fires Kafka events for portfolio snapshots, XIRR recalculation, expense aggregation, and PDF/Excel report generation. (*Planned – Phase 3*)
- Tech: Java 21, Spring Boot 3.x, Kafka, Spring Cloud Gateway, Loki, Grafana, Docker, GitHub Actions.

CAS Parser — Python & Spring Boot

- CAS ingestion via email: user forwards their CAMS/KFintech statement email (with predefined password) to a dedicated service inbox (hi@trails.click); a cron job polls every 5 minutes, extracts the PDF and password, parses holdings using PyPDF2 (Python), builds a consolidated portfolio JSON, and publishes it to a Kafka topic — consumed by the Portfolio service to persist to PostgreSQL.
- The parsed portfolio JSON feeds directly into the Money Trails Portfolio service via Kafka.

EDUCATION

Master of Computer Applications (MCA) 2015 – 2018

Heritage Institute of Technology, Kolkata | MAKAUT | CGPA: 8.36 / 10

Bachelor of Science in Computer Science (Honours) 2011 – 2014

Michael Madhusudan Memorial College, Durgapur | Burdwan University | 71.3%

ADDITIONAL INFORMATION

Languages spoken: English, Hindi

Open to: Java backend engineering roles — across backend, platform, and distributed systems.

Portfolio & Projects: soumen.trails.click | GitHub: github.com/me-soumen